

Acoustic Tagger-Assisted Inference and Audio-Contrastive Adaptation for Audio-Dependent Question Answering

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Abstract—This paper presents our first-place system for DCASE 2026 Task 5, Audio-Dependent Question Answering (ADQA), a multiple-choice benchmark designed to assess whether audio-language systems ground their answers in the input audio rather than relying on textual priors. Our system is built around MOSS-Audio-8B-Thinking and emphasizes training-free audio-aware inference using calibrated answer-letter prompting, rule-routed acoustic tagger assistance, and permutation-based ensemble voting. We additionally investigated GDPO-based LoRA adaptation with audio-contrastive hard examples selected by comparing original-audio and silent-control option scores. On the official ADQA-Bench evaluation set, our training-free tagger-assisted MOSS system achieved 57.30% Top-1 accuracy and ranked first among lightweight submissions. Although the GDPO-adapted MOSS system achieved a slightly lower accuracy of 57.07%, it contributed complementary predictions for ensemble construction. The final mixed ensemble, combining permutation-based MOSS predictions with Qwen-based predictions, achieved 58.33% Top-1 accuracy and ranked first overall. These results demonstrate that training-free question-aware acoustic evidence can form a competitive lightweight ADQA configuration, while diverse prediction sources can be practically exploited within the final ensemble.

Index Terms—Audio question answering, audio-language models, audio-dependent reasoning, acoustic tagging, ensemble inference

1. INTRODUCTION

Audio question answering (AQA) requires a model to infer an answer from both an audio signal and a natural-language question, making it a query-conditioned audio understanding task [1], [2]. Unlike audio captioning or closed-set audio classification, the relevant acoustic evidence is determined by the question and may involve speech content, speaker attributes, music characteristics, environmental sounds, temporal structure, or their interactions. This query-conditioned nature makes AQA a useful testbed for large audio-language models (LALMs), but recent studies show that reliable audio-grounded reasoning remains difficult, especially when the answer requires fine-grained temporal evidence or when models can exploit textual priors without using the input audio [3], [4].

DCASE 2026 Task 5 addresses this issue through Audio-Dependent Question Answering (ADQA), a multiple-choice benchmark designed to evaluate whether answers are grounded in the provided audio rather than inferred from the question and answer choices alone [5]. The task uses Audio-Dependency Filtering (ADF), including silent-audio filtering, common-sense checks, soft text-shortcut filtering, and manual verification, to construct examples whose answers should depend on the input signal. Under this hard-filtered setting, a strong system should not merely be a capable language reasoner. It should remain sensitive to the presence, absence, and discriminative content of the audio while producing stable multiple-choice predictions.

Recent LALMs have extended language models toward general audio understanding by coupling audio encoders with instruction-following or reasoning-oriented language models, covering speech, environmental sounds, music, and open-ended audio interaction [6]–[9]. However, strong general audio-language capability does not necessarily guarantee audio-dependent decision making in multiple-choice settings, where models may still exploit textual priors or option-position biases.

Motivated by this gap, we study a task-specific grounding framework built on MOSS-Audio-8B-Thinking [10], rather than proposing a new audio-language backbone. Our primary focus is training-free, question-aware acoustic evidence injection: a rule-routed tagger selects specialized analyzers according to the question and answer choices and converts their outputs into weak textual evidence under a shared answer-letter interface. We additionally use original-audio versus silent-control option scores to identify grounding-sensitive examples for exploratory Group Reward-Decoupled Normalization Policy Optimization (GDPO)-based LoRA adaptation. Prompt calibration, choice permutation, and mixed-model fusion are treated as robustness mechanisms rather than new modeling contributions.

Under the official DCASE 2026 Task 5 setting, the training-free tagger-assisted MOSS configuration achieved 57.30% Top-1 accuracy and ranked first among lightweight submissions, while the final mixed ensemble achieved 58.33% and ranked first overall. The adapted branch remained competitive but did not outperform the training-free configuration as a standalone system, instead serving as a complementary prediction source for ensemble construction. Because the evaluated configurations differ in multiple components, we interpret these results as system-level comparisons rather than controlled estimates of individual component effects.

2. METHOD

2.1. Calibrated answer-letter prompting

All submitted systems use a calibrated multiple-choice prompt as the shared interface between the audio-language model and the ADQA task. Because large language models can be sensitive to prompt format, answer verbalization, and option-position bias, we standardize the output space before applying acoustic tagging, adaptation, or ensembling [11], [12]. As shown in Figure 1, the prompt explicitly instructs the model to rely on the input audio rather than the question or answer choices alone, and constrains the output to a single valid answer letter. In preliminary prompt comparisons, this answer-letter format was more stable than free-form answer-text generation. We therefore use the same compact interface for direct training-free inference, likelihood-based diagnostics, RL rollouts, and choice-permutation aggregation, which simplifies post-processing and enables consistent option-level scoring.

The same prompt is used for direct training-free inference with MOSS-Audio-8B-Thinking, original-audio versus silent-control diagnostics, choice-permutation aggregation, and GDPO-based LoRA adaptation. When rule-routed acoustic evidence is available, it is

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You are an expert in audio understanding.
IMPORTANT:
- You MUST listen to the audio carefully before answering.
- You MUST rely on the audio content to answer the
question.
- Do NOT answer based only on the text of the question or
choices.
- Answer with ONLY one letter: {valid letters}.
Question: {question}
Choices:
{choice lines}
Answer:

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Fig. 1: Prompt template used as the shared answer-letter interface for direct inference, audio-contrastive log-probability diagnostics, RL rollouts, and choice-permutation scoring.

inserted above the question without changing the answer-letter format, allowing routed and non-routed predictions to be aggregated consistently.

2.2. Question-aware acoustic tagger-assisted inference

The main training-free component of our system is a question-aware acoustic tagger that injects weak auxiliary evidence into the audio-language model without modifying its parameters. Since ADQA questions can depend on heterogeneous acoustic cues, such as speech timing, prosody, speaker changes, music, or background scenes, a single end-to-end LALM may not reliably extract all relevant evidence in a multiple-choice setting. We therefore use a lightweight routing module to select an appropriate auxiliary analyzer for each question.

Our tagger-assisted inference is related to modular audio query systems that route user queries to specialized audio expert models and use their outputs for language-model-based response generation [19]. In contrast, our method is designed for hard-filtered multiple-choice ADQA: the tagger is not an independent answering module, but injects short weak-evidence snippets into a shared answer-letter interface.

Given a question and its answer choices, the router normalizes the text and applies high-precision lexical and phrase patterns to identify the required acoustic cue. Silence or duration cues activate VAD analysis, stress or intonation cues activate prosody analysis, speaker-change cues activate diarization, same-speaker queries activate speaker verification, and music or background-scene cues activate CLAP-based matching. The router uses only the surface form of the question and choices, without labels or evaluation annotations. When multiple routes are activated, a fixed priority order favors more specialized speaker- and scene-related routes. If no reliable rule fires, the example enters the no-auxiliary path; thus, fallback is determined by rule coverage rather than model confidence or the correct answer.

The acoustic tagger is not used as an independent answer predictor, feature-fusion module, or separate voting branch. Instead, each activated analyzer produces a short natural-language evidence block, which is inserted above the question while preserving the calibrated answer-letter interface in Section 2.1. The final answer is always selected by MOSS-Audio under the same multiple-choice prompt. This keeps the prediction format consistent across routed and non-routed examples while allowing the model to condition on interpretable acoustic cues.

Table 1 summarizes the auxiliary analyzers and injected evidence. Heterogeneous analyzer outputs are reduced to short task-relevant statistics, such as speech/pause timing, speaker-turn counts, or option-wise similarity scores, and verbalized as weak auxiliary evidence rather than answer predictions. The prompt instructs the model to prioritize the audio itself when this evidence conflicts with its direct

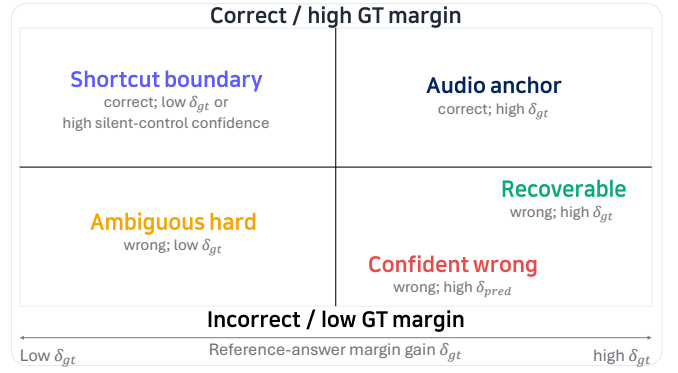


Fig. 2: Conceptual boundary space for audio-contrastive hard-example mining. The horizontal axis represents the reference-answer margin gain δ_{gt} , while the vertical axis represents the ground-truth margin $m_i^{\text{audio}}(y_i)$ under the original audio, with zero separating correct and incorrect predictions. Confident-wrong cases are identified among incorrect examples using the additional δ_{pred} criterion.

audio understanding, reducing the risk that noisy analyzer outputs override the model’s own perception.

For unrouted examples, the training-free MOSS branch uses a fixed no-auxiliary prompt policy based on predefined text-based criteria over the question and answer choices. Structured semantic cases receive a reasoning-ban variant, whereas audio-perceptual or ambiguous cases retain the calibrated base prompt. This policy is determined before inference and does not depend on model confidence or ground-truth labels. The GDPO-adapted branch instead uses the calibrated base prompt without the reasoning-ban instruction, which performed better during development.

2.3. Audio-contrastive hard-example adaptation

In addition to the training-free acoustic tagger-assisted branch, we explore a GDPO-adapted MOSS branch using audio-contrastive hard-example mining. This branch is intended to provide a complementary prediction source rather than replace the training-free system. We therefore select training examples by comparing the model’s option scores under the original audio and a silent control signal.

Let $x_i = (a_i, q_i, C_i, y_i)$ denote an example with audio a_i , question q_i , answer choices C_i , and reference answer y_i . We also define a silent control audio a_i^0 . For each condition $r \in \{\text{audio}, \text{silent}\}$, we compute option-level scores

$$s_i^r(k) = \log P_M(k \mid a_i^r, q_i, C_i), \quad (1)$$

where k denotes an answer option. For a target answer t , we define the option margin as $m_i^r(t) = s_i^r(t) - \max_{k \neq t} s_i^r(k)$, and the audio-contrastive margin gain as $\delta_i(t) = m_i^{\text{audio}}(t) - m_i^{\text{silent}}(t)$. We further denote the gain for the reference answer as $\delta_{gt} = \delta_i(y_i)$, and the gain for the model-selected audio-conditioned answer as $\delta_{pred} = \delta_i(\hat{y}_i^{\text{audio}})$. Figure 2 illustrates this conceptual space using δ_{gt} as the primary horizontal quantity and $m_i^{\text{audio}}(y_i)$ to separate correct from incorrect predictions; δ_{pred} is used as an additional criterion for confident-wrong examples.

Based on these diagnostics, correct examples with high δ_{gt} are treated as audio anchors, whereas correct examples with low δ_{gt} or high silent-control confidence are treated as shortcut-boundary cases. Incorrect examples are further separated into recoverable, confident-wrong, and ambiguous-hard groups: recoverable cases have high δ_{gt} , confident-wrong cases have high δ_{pred} , and ambiguous-hard cases have low δ_{gt} . Correct examples not assigned to the targeted audio-anchor or shortcut-boundary groups are retained as coverage examples.

Table 1: Auxiliary analyzers and feature extractors used by the acoustic tagger-assisted inference module.

Route / acoustic cue	Auxiliary analyzer	Injected evidence
VAD / silence / duration	librosa-based signal features [13]	Speech/silence timing and pauses.
Prosody / stress / intonation	Whisper with acoustic features [14]	Word timing and prosodic cues.
Speaker turn changes	pyannote speaker diarization [15]	Speaker segments and turn count.
Same-speaker verification	ECAPA-TDNN embeddings [16], [17]	Speaker similarity scores.
Music detail	CLAP audio–text matching [18]	Music-choice similarity scores.
Background scene	CLAP audio–text matching [18]	Background-scene similarity scores.

Table 2: Audio-contrastive group composition of AudioMCQ Strong-AC and the final RL training dataset.

Group	AudioMCQ Strong-AC	RL rows
Coverage	79,957	4,096 (32%)
Ambiguous hard	34,908	3,200 (25%)
Audio anchor	32,746	1,920 (15%)
Shortcut boundary	11,016	1,024 (8%)
Confident wrong	7,992	1,920 (15%)
Recoverable	922	640 (5%)
Total	167,541	12,800 (100%)

Thresholds are determined from empirical quantiles of the diagnostic statistics rather than tuned on the development set.

The final training dataset contains 12,800 rows selected from 167,541 AudioMCQ Strong-AC examples [4], as summarized in Table 2. We use group-level quotas and sample within each group approximately in proportion to the square root of each source-dataset-by-question-type cell size, with repeat caps to avoid excessive duplication. This balances shortcut-prone and acoustically confusing cases with coverage-preserving and audio-grounded examples.

For RL variants, we also apply choice-layout calibration during data construction to reduce answer-position imbalance. Based on preliminary inference analysis, the target answer position is sampled with weights A/B/C/D = 1.0/1.0/1.0/1.3, yielding a final target distribution of 22.77/23.70/23.00/30.53%.

We adapt MOSS-Audio-8B-Thinking with LoRA adapters [20] using the mined training reservoir. Each row contains the audio and calibrated multiple-choice prompt, while the reference answer letter is used only for external reward computation. During training, the policy samples multiple completions for each prompt, and the reward combines answer correctness and output validity:

$$R = R_{\text{acc}} + 0.5R_{\text{fmt}}, \quad (2)$$

where R_{acc} rewards a completion whose final answer letter matches the reference answer, and R_{fmt} rewards a parseable answer-letter format. We use a group-relative policy optimization objective with GDPO-style reward scaling across sampled completions [21], [22]. The representative GDPO run uses LoRA rank 16 and alpha 32 on all linear modules, eight generations per prompt, 400 optimization steps, a learning rate of 7×10^{-7} , and maximum completion length 512.

2.4. Permutation-based ensemble and mixed-model fusion

The final component of our system is prediction-level aggregation. Prior work has shown that LLM multiple-choice predictions can be highly sensitive to option order and option-ID biases [23], [24]. We therefore use ensembling to reduce option-order sensitivity and exploit prediction diversity across the training-free MOSS branch, the GDPO-adapted MOSS branch, and the Qwen-based branch.

For the MOSS-only ensemble, we generate multiple permuted views of each example by reordering the answer choices while keeping the

audio and question unchanged. Each predicted answer letter is mapped back to the original choice index, and predictions from the training-free and GDPO-adapted MOSS branches are aggregated by majority voting. This procedure stabilizes multiple-choice predictions without changing the underlying model parameters.

For the final mixed ensemble, we additionally include Qwen3-Omni-30B [25] as an inference-only prediction source. The Qwen branch is not further trained and uses the same answer-letter interface, choice-permutation procedure, and option-level log-probability extraction as the MOSS branches. For each valid view, option log-probabilities are first mapped back to the original choice order. Let $\mathcal{V}_M$ and $\mathcal{V}_Q$ denote the valid MOSS and Qwen views, respectively, and let $\ell_{v,k}$ be the log probability assigned to option k in view v . The final prediction is obtained by weighted log-probability fusion:

$$\hat{y} = \arg \max_k \left(\frac{1}{|\mathcal{V}_M|} \sum_{v \in \mathcal{V}_M} \ell_{v,k} + \lambda \frac{1}{|\mathcal{V}_Q|} \sum_{v \in \mathcal{V}_Q} \ell_{v,k} \right), \quad (3)$$

where λ controls the relative contribution of the Qwen branch. We use $\lambda = 0.46$ in the submitted system, selected from development-set performance.

Because all branches share the same answer-letter interface and are mapped back to the original choice order before aggregation, predictions can be combined consistently across model families and choice permutations.

3. EXPERIMENTS

3.1. Dataset and evaluation

We evaluate under the official DCASE 2026 Task 5 Audio-Dependent Question Answering (ADQA) setting [5]. The development set contains 1,607 examples and is used for system selection, including prompt, routing, adaptation, permutation-ensemble, and mixed-model-fusion choices. GDPO training itself uses the separate AudioMCQ Strong-AC reservoir described in Section 2.3. The official ADQA-Bench evaluation set is used only for final evaluation and not for system selection. All results are reported as Top-1 multiple-choice accuracy after mapping predicted answer letters back to the corresponding answer text required by the official evaluation format.

3.2. System configurations

Table 3 summarizes four official system configurations. All configurations use the calibrated answer-letter interface in Section 2.1. The MOSS-based configurations use MOSS-Audio-8B-Thinking, while the final mixed ensemble additionally includes Qwen3-Omni-30B as an inference-only branch.

S1 is the training-free lightweight submission. It uses the pretrained MOSS-Audio model with acoustic tagger-assisted inference and the no-auxiliary prompt selector described in Section 2.2. S2 uses the GDPO-adapted MOSS branch from Section 2.3 with acoustic tagger assistance, but does not use the no-auxiliary prompt selector; instead, it

Table 3: System-level comparison on the development and official evaluation sets. Development accuracy is computed over 1,607 examples. I1–I2 are internal configurations and S1–S4 are official system configurations. Rows differ in multiple components and should not be interpreted as factorial ablations. M/Q denotes the number of MOSS/Qwen prediction views, respectively.

ID	Calib. prompt	Acoustic tagger	No-aux router	GDPO	Views (M/Q)	Aggregation	Dev Acc. (%)	Eval Acc. (%)
<i>Internal development configurations</i>								
I1	✓	–	–	–	1/0	–	65.837	–
I2	✓	–	–	✓	1/0	–	67.330	–
<i>Official system configurations</i>								
S1	✓	✓	✓	–	1/0	–	69.633	57.30
S2	✓	✓	–	✓	1/0	–	69.695	57.07
S3	✓	✓	mixed	mixed	10/0	Hard vote	70.006	58.10
S4	✓	✓	mixed	mixed	4/2	Weighted log-prob.	70.504	58.33

uses the calibrated base prompt without the reasoning-ban instruction, which performed better during development. Therefore, S1 and S2 should be interpreted as different single-system configurations rather than a fully controlled ablation of GDPO alone.

S3 is the MOSS-only permutation ensemble. It combines five choice-permutation views from S1 and five views from S2, and aggregates the ten MOSS predictions by hard voting. S4 is the final mixed MOSS–Qwen ensemble. It combines two training-free MOSS views, two GDPO-adapted MOSS views, and two Qwen3-Omni-30B views using the weighted log-probability fusion in Section 2.4. Qwen3-Omni-30B is not fine-tuned on the ADQA training data.

3.3. Development and official evaluation results

Table 3 summarizes the development and official ADQA-Bench results. Because the evaluated rows differ in more than one factor, we treat Table 3 as a system-level comparison rather than a factorial ablation. In particular, S1 and S2 differ in both adaptation and no-auxiliary prompt policy, whereas S3 and S4 differ in model composition, number of prediction views, and aggregation method.

I1 and I2 provide a pre-tagger comparison of the base and adapted configurations. I1, which uses the calibrated answer-letter prompt, achieved 65.837% development accuracy, whereas I2, which additionally includes GDPO adaptation and choice-layout calibration, achieved 67.330%. Because these two changes are coupled in I2, the 1.493-point difference should not be attributed to GDPO alone.

Among the single-MOSS configurations, S1 and S2 achieved nearly identical development accuracy, at 69.633% and 69.695%, respectively. On the official evaluation set, S1 achieved 57.30% Top-1 accuracy and ranked first among lightweight submissions, while S2 achieved 57.07%. Because the two configurations differ in both adaptation and no-auxiliary prompt policy, this small difference does not isolate either factor.

The ensemble configurations achieved the highest observed accuracies. S3 reached 70.006% on development and 58.10% on the official evaluation set, ranking second overall, while S4 reached 70.504% and 58.33%, respectively, ranking first overall. The improvement of S3 over the single-MOSS configurations reflects the combined effects of choice permutation, branch diversity, and aggregation. Likewise, because S3 and S4 differ in model composition, view count, and aggregation method, their 0.23-point official difference should not be interpreted as an isolated effect of Qwen fusion.

Overall, the results establish strong system-level performance for the training-free lightweight configuration and the final ensembles. The GDPO-adapted branch remained competitive as a complementary prediction source, but the available comparisons do not isolate the individual effects of acoustic tagging, adaptation, choice permutation, or mixed-model fusion.

3.4. Scope and limitations

Several limitations remain. The calibrated prompt was selected from preliminary comparisons without a complete instruction-level ablation, and the audio-contrastive miner was not compared with size-matched random or conventional hard-example sampling. Because statistical uncertainty was not estimated, small differences between configurations should not be over-interpreted. The tagger may also fail through lexical misrouting, analyzer errors, or domain mismatch; the no-route fallback and audio-priority instruction mitigate but do not eliminate these risks. Controlled component ablations, uncertainty estimates, and direct comparisons of mining strategies remain important future work.

4. CONCLUSION

We presented a system-level study of training-free question-aware acoustic evidence injection, audio-contrastive GDPO adaptation, and prediction-level aggregation for DCASE 2026 Task 5 ADQA. Our primary lightweight configuration uses rule-routed acoustic analyzers to provide weak question-specific evidence while retaining the audio-language model as the final decision maker, whereas the GDPO-adapted branch is treated as an exploratory complementary prediction source.

On the official ADQA-Bench evaluation set, the training-free tagger-assisted MOSS configuration achieved 57.30% Top-1 accuracy and ranked first among lightweight submissions, while the final mixed ensemble achieved 58.33% and ranked first overall. These results demonstrate strong practical system-level performance, but the evaluated configurations do not isolate the individual effects of adaptation, choice permutation, or Qwen fusion.

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